Tobias Mwangi-Hartley

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PROFESSIONAL SUMMARY

Data scientist with experience building and evaluating forecasting, churn and experimentation solutions using Python, SQL and cloud-based data tools. Developed scheduled and batch-scored machine learning workflows, engineered model features, and introduced pytest coverage for forecasting feature code. MSc in Statistics with a dissertation on Bayesian hierarchical models; now seeking to develop further in machine learning engineering.

CORE SKILLS

Programming and analysis: Python, SQL, R
Machine learning: LightGBM, XGBoost, scikit-learn, pandas, SHAP, feature engineering, forecasting, churn modelling
Data and cloud tools: BigQuery, Vertex AI Workbench, dbt, Looker
Engineering and applications: pytest, Docker, Streamlit, scheduled notebooks, batch scoring
Statistics and experimentation: Bayesian hierarchical models, experimental design, sequential A/B testing, statsmodels
Additional learning: PyTorch, torchvision (course project)

PROFESSIONAL EXPERIENCE

Data Scientist *Thistlebox | 2022-Present*

- Built a six-week-ahead weekly recipe-demand forecasting workflow using LightGBM and calendar, price and menu-position features. The scheduled Vertex AI Workbench notebook writes forecasts to BigQuery for operational use in Looker.
- Improved forecast WAPE from 23% to 14%; the project estimated a 9% reduction in food waste.
- Developed a weekly XGBoost churn-scoring workflow in BigQuery and used top-decile scores to target retention offers. A 10% holdout validation showed churn was 2.1 percentage points lower in the treated group.
- Built a Streamlit application for planning and analysing sequential A/B tests, used by approximately 25 people in product and growth. Wrote the app and Dockerfile; the data platform team handles deployment to Cloud Run.
- Began writing pytest unit tests for forecasting feature code in 2025; coverage is approximately 60%.
- Delivered the internal training session “Stats for PMs” three times.

Junior Data Analyst *Firth Energy Retail | 2020-2022*

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

ADDITIONAL PROJECT

Image Classification Course Project *2025*

- Fine-tuned a small CNN in PyTorch on a public food-images dataset as part of an online deep learning course; achieved approximately 80% accuracy.
- Tools: PyTorch, torchvision, Google Colab.

EDUCATION

MSc in Statistics
Dissertation: Bayesian hierarchical models